A Constructive Introduction to Quantum Field Theory

Lecture 3 — Stochastic observables, the Polchinski flow, FBSDEs

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Notes: Lecture 3, §§4–10

Where we are

So far. A Euclidean field is a measure on $\mathcal{S}'(\mathcal{M})$; the GFF μ_m is its Gaussian part; local operators are functions of φ. Wick powers $:\varphi^n:$ are random distributions iff $n(d-2) < d$, and not even tight otherwise.

Today. Index everything by a resolution $t \in [0, \infty)$.

1. The GFF as the terminal value of a martingale of smooth fields; observables = martingales.
2. Interactions: the Polchinski equation and its FBSDE; the force is an observable.
3. The FBSDE in force coordinates $\Rightarrow$ the Wetterich equation.
4. Variational structure: Boué–Dupuis, HJB, infinite volume.
5. Additional material: products and OPE, flatness, counterterms for Φ^4_3, DS equations at every scale.

The idea

- Wilson–Kadanoff: physics is organised scale by scale. The measure-on-$\mathcal{S}'$ picture hides this, and forces singular objects from the start.
- Observe the field at resolution t: a smooth function φ_t. Information at resolution t: $\mathcal{F}_t = \sigma(\varphi_s : s \le t)$.
- A physical quantity is not a random variable but a consistent family of finite-resolution estimates, i.e. a martingale.

Slogan

Observables are martingales in the filtration of finite-resolution measurements. They need not have a terminal value.

Caveat. Statements marked (formal) are exact in finite dimensions, with a terminal resolution $T < \infty$; the limits $T \to \infty$, $L \to \infty$ are the real problems.

Bailleul–Chevyrev–G. (in prep.); Polchinski 1984; Bauerschmidt–Bodineau–Dagallier 2024

Setting: covariance profiles and observables

Covariance profiles

Definition

A covariance profile is a family $(Q_t)_{t\geq 0}$ of symmetric, translation invariant operators with $Q_0 = 0$, $\dot{Q}_t := \partial_t Q_t \geq 0$, $Q_t \to C = (m^2 - \Delta)^{-1}$, and smooth kernels for $t < \infty$.

Heat profile:

$$Q_t = C\,\mathrm{e}^{-A/t^2}, \qquad \dot{Q}_t = \frac{2}{t^3}\mathrm{e}^{-A/t^2}, \qquad \dot{Q}_t(x-y) = \frac{2t^{d-3}\mathrm{e}^{-m^2/t^2}}{(4\pi)^{d/2}}\,\mathrm{e}^{-t^2|x-y|^2/4} \text{ on } \mathbb{R}^d.$$

$\dot{Q}_t$ lives on distances $\lesssim 1/t$; Q_t keeps the modes $|k| \lesssim t$: t is a momentum scale.

Others: sharp Fourier cutoff $\mathbb{1}_{|k|\leq t}(|k|^2 + m^2)^{-1}$; finite range decompositions, $\dot{Q}_t(x,y) = 0$ for $|x-y| \geq 1/t$.

Notation: $c_t = Q_t(x,x)$, $Q_{T,t} = Q_T - Q_t$. For $d \geq 2$, $c_t \to \infty$ ($\frac{1}{2\pi}\log t$ in $d = 2$, $\asymp t^{d-2}$ in $d \geq 3$).

The field at resolution t

W cylindrical Brownian motion on $L^2(\mathcal{M})$,

$$X_t := \int_0^t \dot{Q}_s^{1/2} \, \mathrm{d}W_s.$$

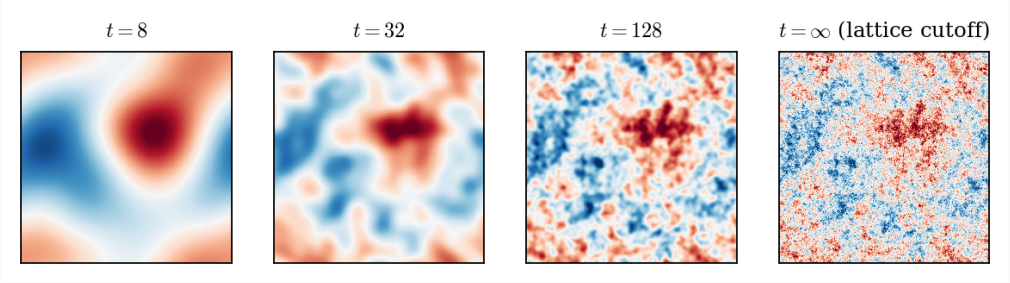
Proposition

X_t is a smooth Gaussian field, $\mathrm{Law}(X_t) = \mu^{Q_t}$*; $(X_t(x))_t$ is a continuous martingale with* $\mathrm{d}\langle X(x), X(y)\rangle_t = \dot{Q}_t(x,y)\,\mathrm{d}t$*; $X_t \to X_\infty$ in H^{-s}_{loc}, $s > \frac{d}{2} - 1$, with* $\mathrm{Law}(X_\infty) = \mu_m$*; $X_T - X_t \perp \mathcal{F}_t$, with law $\mu^{Q_{T,t}}$.*

The GFF is the terminal value of a Gaussian martingale of smooth fields.

Proof (torus). $X_t = \sum_n \big(\int_0^t \dot{q}_n(s)^{1/2} \mathrm{d}B_s^n\big) e_n$: independent Gaussian martingales, variance $q_n(t) \uparrow \lambda_n^{-1}$; then as in Lecture 2. □

The field at resolution t: a picture



Heat profile on $\mathbb{T}^2$, $m = 4$: $t = 8, 32, 128$ and the lattice cutoff, same white noise.

The martingale reveals finer and finer details; the large scales are fixed at low resolution.

Stochastic observables

$\mathbb{E}_t = \mathbb{E}[\,\cdot\,|\mathcal{F}_t]$.

Definition (Bailleul–Chevyrev–G.)

An observable is an $(\mathcal{F}_t)$-martingale $O = (O_t)_{t\geq 0}$. Its expectation is $\langle O\rangle = \mathbb{E}O_t$. An observable field is a family $(O(x))_{x\in\mathcal{M}}$.

- $Y \in L^1(\mathcal{F}_\infty)$ gives a closed observable $O_t = \mathbb{E}_t Y$.
- The point: observables need not be closed. $X_t(x)$, $d \geq 2$, diverges, yet is a perfectly good family of estimates.

Asymptotic uniqueness

(a) $\|O^1_t - O^2_t\|_{L^1} \to 0 \Rightarrow O^1 = O^2$. (b) N adapted, $\|N_t - O_t\|_{L^1} \to 0 \Rightarrow O_s = \lim_t \mathbb{E}_s N_t$.

Proof. $\mathbb{E}_s$ is an L^1 contraction. □

Germs and locality

An observable is determined by its asymptotics.

- A germ of O: functionals $\mathring{O}_t$ of smooth fields with $\|O_t - \mathring{O}_t(X_t)\|_{L^1} \to 0$. A germ determines O.
- $O(x)$ is local if it has germs

$$\mathring{O}_t(X_t)(x) = \mathring{\mathcal{O}}_t\big(X_t(x), \nabla X_t(x), \dots, \nabla^k X_t(x)\big).$$

- Germs are not unique: this freedom is the freedom of choosing counterterms.

Example. $:X_t^n:(x) = H_n(X_t(x); c_t)$ is a local observable in every d (more later), although $:\varphi^n:$ does not exist as a random distribution when $n(d-2) \geq d$.

Interactions: the Polchinski equation and FBSDEs

Effective potentials

Finite volume $\mathbb{T}_L^d$, terminal resolution $T < \infty$ (UV cutoff), interaction V_T bounded below, e.g.

$$V_T(\varphi) = \int_{\mathbb{T}_L^d} \big[\lambda H_4(\varphi; c_T) + \gamma_T H_2(\varphi; c_T)\big], \qquad \nu_T(\mathrm{d}\varphi) = Z_T^{-1} \mathrm{e}^{-V_T(\varphi)} \mu^{Q_T}(\mathrm{d}\varphi).$$

Definition (Wilson effective action)

$$\mathrm{e}^{-V_t(\varphi)} := \int \mathrm{e}^{-V_T(\varphi+\zeta)} \, \mu^{Q_{T,t}}(\mathrm{d}\zeta), \qquad t \in [0, T].$$

The interaction felt at resolution t once the scales $[t, T]$ are integrated out; $V_0(0) = -\log Z_T$ is the free energy.

Marginals and stochastic localisation

Proposition (Marginals)

$\mathrm{d}\mathbb{P}/\mathrm{d}\mathbb{P}_0 = Z_T^{-1}\mathrm{e}^{-V_T(X_T)}$. *Then* $\mathrm{Law}_{\mathbb{P}}(X_t) = \nu_t \propto \mathrm{e}^{-V_t}\mu^{Q_t}$, *and* $\mathrm{Law}_{\mathbb{P}}(X_T|\mathcal{F}_t) = \mu_t^{X_t}$, *the tilt of* $\mu^{Q_{T,t}}(\cdot - X_t)$ *by* e^{-V_T}.

- At every resolution the field is again a Gibbs measure: reference Q_t, potential V_t.
- The measure-valued martingale $t \mapsto \mu_t^{X_t}$ goes from ν_T ($t = 0$) to δ_{X_T} ($t = T$): a stochastic localisation of ν_T (Eldan), adapted to the covariance profile.

Proof of the marginals. Under $\mathbb{P}_0$, $X_T = X_t + (X_T - X_t)$ with independent increment $\sim \mu^{Q_{T,t}}$, so

$$\mathbb{E}_{\mathbb{P}}[G(X_t)] = Z_T^{-1}\mathbb{E}_0\big[G(X_t)\mathrm{e}^{-V_T(X_T)}\big] = Z_T^{-1}\mathbb{E}_0\big[G(X_t)\mathrm{e}^{-V_t(X_t)}\big]. \qquad \square$$

The Polchinski equation

Theorem (Polchinski 1984)

(formal)

$$\partial_t V_t = \tfrac{1}{2}\langle DV_t, \dot{Q}_t\, DV_t\rangle - \tfrac{1}{2}\operatorname{Tr}\big(\dot{Q}_t\, D^2 V_t\big), \qquad V_T \text{ given.}$$

Proof. $u_t = \mathrm{e}^{-V_t} = \mathrm{e}^{\frac{1}{2}\Delta_{Q_{T,t}}} u_T$, $\Delta_Q = \operatorname{Tr}(QD^2)$, solves the backward heat equation $\partial_t u_t = -\frac{1}{2}\operatorname{Tr}(\dot{Q}_t D^2 u_t)$. Insert $D^2 u = (DV \otimes DV - D^2 V)u$. □

Nonlinear term: generates new, nonlocal interactions: trees.

Trace term: contracts pairs of fields with the shell covariance $\dot{Q}_t\, \mathrm{d}t$: loops.

A backward Hamilton–Jacobi equation; $u = \mathrm{e}^{-V}$ is a Cole–Hopf transform.

Theorem (FBSDE for the interacting field)

(formal) Under $\mathbb{P}$, the interacting field at resolution t $\Phi_t := X_t$ solves

$$\mathrm{d}\Phi_t = \dot{Q}_t F_t \,\mathrm{d}t + \dot{Q}_t^{1/2}\,\mathrm{d}W_t^{\mathbb{P}}, \qquad F_t = -DV_t(\Phi_t) = \mathbb{E}_t^{\mathbb{P}}\big[-DV_T(\Phi_T)\big].$$

The effective force F is a $\mathbb{P}$-martingale: an observable field of the interacting theory.

Proof. Density process $M_t = Z_T^{-1}\mathrm{e}^{-V_t(X_t)}$. Itô + Polchinski: no drift, $\mathrm{d}M_t = -M_t\langle DV_t(X_t), \dot{Q}_t^{1/2}\mathrm{d}W_t\rangle$. Girsanov: $W_t^{\mathbb{P}} = W_t + \int_0^t \dot{Q}_s^{1/2} DV_s(X_s)\,\mathrm{d}s$. Finally $DV_t(\phi) = \mu_t^{\phi}(DV_T)$. □

A forward–backward system

Read as an equation for an unknown pair (Φ, F):

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t = \mathbb{E}_t\big[-DV_T(\Phi_T)\big], \qquad t \in [0,T].$$

- The field is driven forward by a force determined backward from the terminal field.
- No Polchinski equation needed; $\nu_T = \mathrm{Law}(\Phi_T)$ from any solution.
- Strategy: solve with renormalised V_T, estimates uniform in T and volume, take limits.
- Φ is not an observable (a semimartingale); observables are built from the noise and from F.

Closed versus local

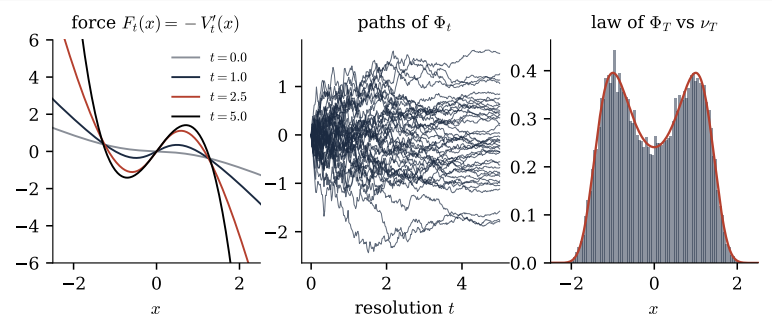
At finite T the force is a closed observable, $F_t = \mathbb{E}_t[F_T]$, and for Φ^4 it is local:

$$F_T(x) = -4\lambda H_3(\Phi_T(x); c_T) - 2\gamma_T H_1(\Phi_T(x); c_T).$$

As $T \to \infty$

F should converge as an observable, but not be closed: its terminal value would be the renormalised cube of the interacting field, which is not a function of Φ_∞. Locality of the limiting force is the crux of the construction.

Zero dimensions



$Q_t = 1 - \mathrm{e}^{-t}$, $T = 5$, $V_T(x) = \frac{1}{2}x^4 - \frac{3}{2}x^2$. Left: $F_t = -V_t'$, almost linear at low resolution, double well as $t \to T$. Centre: FBSDE paths split between the wells (stochastic localisation). Right: histogram of Φ_T vs. the density of ν_T.

Observables of the interacting theory

For G a functional of the terminal field, $O_t = \mathbb{E}_t^{\mathbb{P}}[G(\Phi_T)] = G_t(\Phi_t)$, $G_t(\phi) = \mu_t^{\phi}(G)$, and

$$\partial_t G_t + \tfrac{1}{2}\operatorname{Tr}\big(\dot{Q}_t D^2 G_t\big) - \langle \dot{Q}_t DV_t, DG_t\rangle = 0, \qquad G_T = G :$$

the backward Kolmogorov equation, a Doob transform of the Gaussian heat equation. Any solution gives an observable, closed at finite T, not necessarily as $T \to \infty$. The force is $G = -DV_T$.

Tangent observables. V_T^θ depending on a coupling, $U_t = \partial_\theta V_t^\theta$:

$$U_t(\Phi_t) = \mathbb{E}_t^\theta\big[\partial_\theta V_T^\theta(\Phi_T)\big], \qquad \partial_\theta \mathbb{E}^{\nu_T^\theta}[G] = -\operatorname{Cov}^{\nu_T^\theta}\big(G, \partial_\theta V_T^\theta\big).$$

E.g. the running $:\varphi^4:$: for $\theta = \lambda$: observables conjugate to the coordinates of theory space.

The force: a Burgers equation

Differentiating Polchinski, $F_t = -DV_t$ solves the backward viscous Burgers equation

$$\partial_t F_t = -DF_t[\dot{Q}_t F_t] - \tfrac{1}{2}\operatorname{Tr}\big(\dot{Q}_t D^2 F_t\big), \qquad F_T = -DV_T.$$

- Cole–Hopf $F = D\log u$ linearises it ($u = \mathrm{e}^{-V}$ solves a heat equation).
- Uses only the force: intensive, while V_t is extensive ($\sim$ volume). Matters in infinite volume.
- Flows for different covariances commute iff DF is symmetric, i.e. F a gradient. Non-gradient forces (general Wilson–Itô diffusions): curvature.

The FBSDE in force coordinates: the Wetterich equation

State and costate

$S_t := Q_T - Q_t$: the covariance still unresolved; $\dot{S}_t = -\dot{Q}_t$, $S_T = 0$. $\mathrm{d}F_t = \mathrm{Y}_t\,\mathrm{d}W_t$.

Pontryagin. Control problem $\mathrm{d}X = \dot{Q}_t^{1/2}(u\,\mathrm{d}t + \mathrm{d}W)$, cost $V_T(X_T) + \frac{1}{2}\int \|u\|^2$. Hamiltonian $\langle p, \dot{Q}_t^{1/2}u\rangle - \frac{1}{2}\|u\|^2$ is state independent $\Rightarrow \mathrm{d}p_t = \mathrm{Y}_t\,\mathrm{d}W_t$, $p_T = -DV_T(X_T)$, $u_t = \dot{Q}_t^{1/2}p_t$.

The field Φ is the state; the force F is the costate.

- $F_t(\cdot)$ is the decoupling field, Burgers its quasilinear PDE.
- Transport term $-DF[\dot{Q}F]$ $\leftarrow$ drift of the state: trees.
  Itô term $-\frac{1}{2}\operatorname{Tr}(\dot{Q}D^2F)$ $\leftarrow$ noise: loops.

Noiseless flow from t to T: the linear shear $\Sigma_t : (x, p) \mapsto (x + S_t p,\ p)$, the characteristics of the transport part.

Removing the predictable drift: the forecast

Proposition

(formal) The conditional mean of the terminal field is

$$\psi_t := \mathbb{E}_t[\Phi_T] = \Phi_t + S_t F_t, \qquad (\psi_t, F_t) = \Sigma_t(\Phi_t, F_t),$$

a martingale with $\mathrm{d}\psi_t = \dot{Q}_t^{1/2}\mathrm{d}W_t + S_t\,\mathrm{d}F_t = \big(\mathbb{1} + S_t DF_t(\Phi_t)\big)\dot{Q}_t^{1/2}\mathrm{d}W_t.$

Proof. F martingale $\Rightarrow \mathbb{E}_t \int_t^T \dot{Q}_s F_s\,\mathrm{d}s = S_t F_t$. In the differential, $\dot{Q}_t F_t\,\mathrm{d}t$ cancels with $\dot{S}_t F_t\,\mathrm{d}t$. □

- $\Phi_t = \psi_t - S_t F_t$: a martingale (the field observable) minus the drift still expected on the unresolved scales.
- Noise of ψ: fresh shell fluctuation + revision of the forecast. The forward–backward coupling moves from the drift to the volatility.

ψ is the interaction picture w.r.t. the classical dynamics.

The FBSDE in forecast coordinates

Invert $x \mapsto \psi = x + S_t F_t(x)$: $x = \psi + S_t \mathcal{F}_t(\psi)$, $\mathcal{F}_t(\psi) := -F_t(x)$. Response and regulator:

$$H_t(\psi) = \big(S_t^{-1} + D\mathcal{F}_t(\psi)\big)^{-1}, \qquad \mathcal{R}_t = S_t^{-1} - Q_T^{-1}, \qquad \dot{\mathcal{R}}_t = S_t^{-1}\dot{Q}_t S_t^{-1} \geq 0.$$

Theorem

(formal)

$$\mathrm{d}\psi_t = H_t(\psi_t)\, S_t^{-1} \dot{Q}_t^{1/2}\, \mathrm{d}W_t, \qquad \mathcal{F}_t(\psi_t) = \mathbb{E}_t\big[DV_T(\psi_T)\big],$$

$\psi_T = \Phi_T$. *No drift; covariance* $\mathcal{C}_t = H_t \dot{\mathcal{R}}_t H_t^*$. *The decoupling field solves the pure diffusion equation*

$$\partial_t \mathcal{F}_t + \tfrac{1}{2}\,\mathrm{Tr}\big(\mathcal{C}_t\, D^2\mathcal{F}_t\big) = 0, \qquad \mathcal{F}_T = DV_T.$$

Proof. $\mathbb{1} + S_t DF_t = H_t S_t^{-1}$; $\mathcal{F}_t(\psi_t)$ is a martingale and ψ has no drift: Itô. □

What the change of coordinates did

Compare with Burgers:

$$\underbrace{\partial_t F = -DF[\dot{Q}F] - \tfrac{1}{2}\,\mathrm{Tr}(\dot{Q}D^2F)}_{\text{field coordinate}} \qquad \longrightarrow \qquad \underbrace{\partial_t \mathcal{F} = -\tfrac{1}{2}\,\mathrm{Tr}(\mathcal{C}\,D^2\mathcal{F})}_{\text{forecast coordinate}}$$

- The transport term is absorbed exactly: dropping the Itô term, $\mathcal{F}_t = DV_T$ does not move. All trees live in the coordinate change $x = \psi + S_t\mathcal{F}_t(\psi)$.
- Loops come only from the quadratic variation $\mathcal{C}_t$ of the forecast; the nonlinearity has moved into its field dependence.
- No gradient structure, no Gaussian convolution used: valid for any FBSDE with a Markov martingale force, whenever the coordinate change is invertible.

Conditional moments

Proposition

(formal) If $F_t = -DV_t$, then

$$H_t(\psi_t) = \mathrm{Cov}_t \coloneqq \mathrm{Cov}^{\mathbb{P}}(\Phi_T \mid \mathcal{F}_t) = S_t - S_t D^2 V_t(\Phi_t) S_t.$$

So $H_t > 0$, $\mathbb{1} + S_t DF_t$ is always invertible, and $H_t \leq S_t$ for convex V_T (Brascamp–Lieb).

The forecast coordinates are the first two conditional moments of the stochastic localisation: $(\psi_t, \mathrm{Cov}_t) = (\mathbb{E}_t \Phi_T,\ \mathrm{Cov}_t\, \Phi_T)$. Since $\mathbb{E}_t[\Phi_T \otimes \Phi_T] = \psi_t \otimes \psi_t + \mathrm{Cov}_t$ is a martingale,

$$\mathrm{d}\langle \psi, \psi \rangle_t = \mathrm{Cov}_t\, \dot{\mathcal{R}}_t\, \mathrm{Cov}_t\ \mathrm{d}t, \qquad \mathrm{d}\,\mathrm{Cov}_t = -\,\mathrm{Cov}_t\, \dot{\mathcal{R}}_t\, \mathrm{Cov}_t\ \mathrm{d}t + \mathrm{d}(\text{mart.}).$$

Gaussian reference: $S_t \dot{\mathcal{R}}_t S_t = \dot{Q}_t$.

The gradient sector

$F_t = -DV_t$: the graph $\{(x, -DV_t(x))\}$ is a Lagrangian submanifold of (field, force) space. The shear Σ_t is symplectic (S_t symmetric), and the image is a graph over ψ (invertibility). Hence

$$\mathcal{F}_t = D\Gamma_t, \qquad H_t = \left(S_t^{-1} + D^2\Gamma_t\right)^{-1}, \qquad \Gamma_T = V_T.$$

Generating function: the Moreau transform

$$V_t(x) = \inf_{\psi}\left\{\Gamma_t(\psi) + \tfrac{1}{2}\langle x - \psi, S_t^{-1}(x-\psi)\rangle\right\},$$

optimality: $x = \psi + S_t D\Gamma_t(\psi)$, $DV_t(x) = D\Gamma_t(\psi)$. Well defined: $\frac{1}{2}\langle x, S_t^{-1}x\rangle - V_t(x)$ has Hessian $S_t^{-1}\operatorname{Cov}_t S_t^{-1} > 0$.

Theorem

(formal)

$$\partial_t \Gamma_t = \tfrac{1}{2} \operatorname{Tr}\Big[\dot{\mathcal{R}}_t \Big(\big(S_t^{-1} + D^2\Gamma_t\big)^{-1} - S_t \Big)\Big], \qquad \Gamma_T = V_T.$$

Equivalently, for the effective average action $\Gamma_t^{\mathrm{EAA}} = \Gamma_t + \tfrac{1}{2}\langle \psi, Q_T^{-1}\psi\rangle$,

$$\partial_t \Gamma_t^{\mathrm{EAA}} = \tfrac{1}{2} \operatorname{Tr}\Big[\dot{\mathcal{R}}_t \big(D^2\Gamma_t^{\mathrm{EAA}} + \mathcal{R}_t\big)^{-1}\Big] + \mathit{const},$$

from the bare action ($t = T$, $\mathcal{R}_T = \infty$) to the 1PI effective action of ν_T ($t = 0$, $\mathcal{R}_0 = 0$).

Proof. $D_v \operatorname{Tr}(\dot{\mathcal{R}} H) = -\operatorname{Tr}(\mathcal{C}\, D^2\mathcal{F})[v]$ by symmetry of $D^3\Gamma$, so the force equation reads $D\big(\partial_t\Gamma - \tfrac{1}{2}\operatorname{Tr}(\dot{\mathcal{R}}H)\big) = 0$. The constant: envelope theorem + Polchinski. □

Wetterich = the backward half of the FBSDE

Every object in the Wetterich equation is an observable of the Polchinski process.

Wetterich	FBSDE
field argument ψ	forecast $\psi_t = \mathbb{E}_t[\Phi_T]$
$D\Gamma_t(\psi_t)$	minus the costate, $\mathbb{E}_t[DV_T(\Phi_T)]$
propagator $(D^2\Gamma_t^{\mathrm{EAA}} + \mathcal{R}_t)^{-1}$	conditional covariance Cov_t
regulator $\mathcal{R}_t = S_t^{-1} - Q_T^{-1}$	precision of the unresolved shell minus the prior's
$\partial_t \Gamma_t$	$\frac{1}{2}\,\mathrm{Tr}[\dot{\mathcal{R}}_t(\mathrm{Cov}_t - S_t)]$

Driven by interacting minus free conditional covariance; one-loop exact, with the full propagator; in the Gaussian case only the constant flows.

Wetterich 1993; Morris 1994; Berges–Tetradis–Wetterich 2002

The regulator of the heat profile

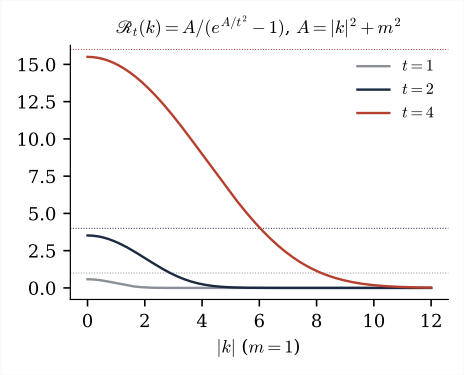
For $T = \infty$, $S_t = C(1 - \mathrm{e}^{-A/t^2})$ and

$$\mathcal{R}_t = \frac{A}{\mathrm{e}^{A/t^2} - 1}.$$

- $\approx t^2$ (a mass term) on the modes $|k| \ll t$;
- exponentially small on $|k| \gg t$.

The exponential regulator of the functional RG.

Other profiles give other regulators, e.g. the mass regulator $\mathcal{R}_t = t^2$.



Variational meaning and the tree level

Constrained free energy. Gibbs variational principle for the conditional law:

$$\Gamma_t(\psi) = \inf\Big\{ \int V_T \,\mathrm{d}\nu + \mathrm{Ent}\big(\nu \mid \mu^{S_t}(\cdot - \psi)\big) : \ \int \varphi \,\mathrm{d}\nu = \psi \Big\} :$$

the free energy of the unresolved shell, constrained to the forecast. V_t is the value function; Γ_t is its Moreau transform.

Tree level. Drop the Itô term: $\Gamma_t = V_T$, and the Moreau transform becomes the Hopf–Lax formula

$$V_t^{\mathrm{tree}}(x) = \inf_y \Big\{ V_T(y) + \tfrac{1}{2}\langle y - x, S_t^{-1}(y - x)\rangle \Big\},$$

which solves $\partial_t V = \frac{1}{2}\langle DV, \dot{Q} DV\rangle$: the saddle point of the Gaussian convolution. The minimiser solves the classical field equation $S_t^{-1}(y - x) + DV_T(y) = 0$ in the shell $[t, T]$; expanding in V_T gives exactly the tree diagrams.

Variational structure

The Boué–Dupuis formula

$\mathbb{H}_a$: adapted drifts with $\mathbb{E}\int_0^T \|u_s\|^2 < \infty$; $X^u = X + \int_0^\cdot \dot{Q}_s^{1/2} u_s \, \mathrm{d}s$.

Theorem (Boué–Dupuis 1998; Polchinski = HJB)

(formal)

$$-\log \mathbb{E}_0\left[\mathrm{e}^{-V_T(X_T)}\right] = V_0(0) = \inf_{u\in\mathbb{H}_a} \mathbb{E}_0\left[V_T(X_T^u) + \frac{1}{2}\int_0^T \|u_s\|^2 \mathrm{d}s\right].$$

Optimal feedback $u_s^* = -\dot{Q}_s^{1/2} DV_s(X_s^{u^*})$, $\mathrm{Law}(X_T^{u^*}) = \nu_T$. *$V_t$ is the value function; Polchinski is the HJB equation.*

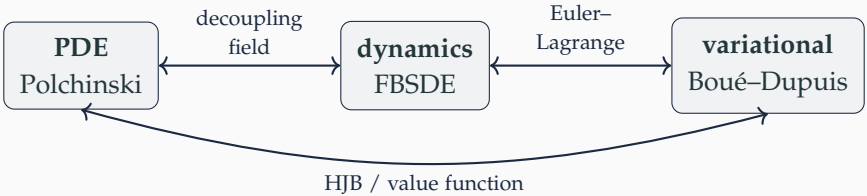
Proof (verification). Itô + Polchinski:

$$\mathbb{E}_0\left[V_T(X_T^u) + \tfrac{1}{2}\int_0^T \|u_t\|^2\right] = V_0(0) + \tfrac{1}{2}\mathbb{E}_0 \int_0^T \|u_t + \dot{Q}_t^{1/2} DV_t(X_t^u)\|^2 \mathrm{d}t. \qquad \square$$

Three faces of one object

Euler–Lagrange. Perturb $u \to u + \varepsilon v$:

$$u_s = -\dot{Q}_s^{1/2}\, \mathbb{E}_s\big[DV_T(X_T^u)\big] = \dot{Q}_s^{1/2} F_s : \quad \text{the FBSDE.}$$



Entropy. $\frac{1}{2}\mathbb{E}_0 \int \|u\|^2 \geq \mathrm{Ent}(\mathrm{Law}(X^u) | \mathrm{Law}(X))$: a pathwise lift of the Gibbs principle $-\log Z_T = \inf_\nu \{ \int V_T \, \mathrm{d}\nu + \mathrm{Ent}(\nu | \mu^{Q_T}) \}$. Useful because adapted drifts can be built scale by scale, as functionals of X.

Renormalisation in the variational picture

Generating functional:

$$\mathcal{W}_T(f) = -\log \int \mathrm{e}^{-f}\,\mathrm{d}\nu_T = \inf_u \mathbb{E}_0\Big[(f+V_T)(X_T^u) + \tfrac{1}{2}\int_0^T \|u_s\|^2\Big] - V_0(0).$$

Bounds uniform in $T \Rightarrow$ tightness, moments, identification of limits.

Φ^4_3. The leading drift $u_s \approx -4\lambda \dot{Q}_s^{1/2} {:}X_s^3{:}$ has divergent cost,

$$\tfrac{1}{2}\mathbb{E}\int_0^T \|4\lambda \dot{Q}_s^{1/2} {:}X_s^3{:}\|^2 \mathrm{d}s = 12\lambda^2 L^3 \int Q_T(z)^4\,\mathrm{d}z \sim T.$$

Change variables $u = -4\lambda \dot{Q}^{1/2} {:}X^3{:} + r$, then paracontrolled changes of variables: the singular terms become explicit polynomials in Wick powers of X, cancelled by the counterterms, and the remainder is coercive. Uniform control in Besov/paracontrolled norms = construction of Φ^4_3.

Barashkov–Gubinelli, Duke 2020; EJP 2021

Infinite volume

On $\mathbb{T}_L^d$, $V_0(0)$ and V_t are extensive ($\sim L^d$). Three intensive objects survive:

(i) **the force.** For $V_T = \int v_T(\varphi)$, on $\mathbb{R}^d$:

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + X_t, \qquad F_t(x) = -\mathbb{E}_t\big[v_T'(\Phi_T(x))\big].$$

Everything is local; infinite-volume limits should solve this. Uniqueness of solutions $\leftrightarrow$ uniqueness of the theory; non-uniqueness expected at phase transitions (Φ_2^4, deep double well).

(ii) **the Burgers flow** of the force;

(iii) **differences of free energies** $\mathcal{W}_T^{(L)}(f)$, f local: bounded as $L \to \infty$; basis of the infinite-volume analysis of Φ_2^4 and $\exp(\Phi)_2$.

Estimates must be local (weighted norms, volume-independent constants): the analogue of "coming down from infinity".

Barashkov–Gubinelli, PMP 2023; Glimm–Jaffe–Spencer 1975

Additional material

Wick observables in every dimension

Recall $\partial_c H_n(a;c) = -\frac{1}{2}\partial_a^2 H_n$, $\partial_a H_n = n H_{n-1}$.

Proposition

For every d, n, x: $:X_t^n:(x) := H_n(X_t(x); c_t)$ is a continuous martingale, $\mathrm{d}:X_t^n:(x) = n:X_t^{n-1}:(x)\,\mathrm{d}X_t(x)$, and

$$\mathbb{E}\big[:X_t^n:(x):X_t^k:(y)\big] = \delta_{nk}\, n!\, Q_t(x,y)^n \xrightarrow{t\to\infty} \delta_{nk}\, n!\, G(x-y)^n \quad (x \neq y).$$

Proof. Itô: the drift is $(\frac{1}{2}\partial_a^2 + \partial_c)H_n\,\dot{c}_t = 0$. □

- Lecture 2: $:\varphi^n:$ exists iff $n(d-2) < d$. As observables, Wick powers exist in every d, unsmeared.
- For $n(d-2) \geq d$, $:X_t^n:(f)$ is unbounded in L^2 and (Paley–Zygmund) not tight: no terminal value; the observable is the whole family.
- The singularity has moved to products at coincident points.

Products of observables

Definition

$O^1 \star O^2$ is the observable with $\|(O^1 \star O^2)_t - O^1_t O^2_t\|_{L^1} \to 0$; unique, and $(O^1 \star O^2)_s = \lim_t \mathbb{E}_s[O^1_t O^2_t]$.

Proposition

If $\mathbb{E} \int_0^\infty |\mathrm{d}\langle O^1, O^2\rangle_s| < \infty$, *then*

$$(O^1 \star O^2)_t = O^1_t O^2_t + \mathbb{E}_t\Big[\int_t^\infty \mathrm{d}\langle O^1, O^2\rangle_s\Big].$$

Proof. $O^1 O^2 - \langle O^1, O^2\rangle$ and $\mathbb{E}_t\langle O^1, O^2\rangle_\infty$ are martingales. □

Products of point evaluations

$\langle \mathbf{X}(x), \mathbf{X}(y) \rangle_t = Q_t(x-y)$ is deterministic. Heat profile, $d \geq 2$: the condition holds iff $x \neq y$, and then

$$(\mathbf{X}(x) \star \mathbf{X}(y))_t = X_t(x)X_t(y) + G(x-y) - Q_t(x-y), \qquad \langle \mathbf{X}(x) \star \mathbf{X}(y) \rangle = G(x-y).$$

The correct two-point function, without ever discussing the regularity of X_∞. For $x = y$: no product, since $c_t \to \infty$.

OPE. At every fixed resolution, as $y \to x$,

$$(\mathbf{X}(x) \star \mathbf{X}(y))_t - G(x-y) \;\longrightarrow\; X_t(x)^2 - c_t = :X_t^2:(x),$$

i.e. $\mathbf{X}(x) \star \mathbf{X}(y) = G(x-y)\mathbb{1} + :\mathbf{X}^2:(x) + o(1)$: the Wick square is the regular part of the product.

The operator product expansion

For $x \neq y$,

$$:\mathbf{X}^n:(x) \star :\mathbf{X}^k:(y) = \sum_{j \leq n \wedge k} j! \binom{n}{j} \binom{k}{j} G(x-y)^j :\mathbf{X}^{n-j}(x)\mathbf{X}^{k-j}(y):.$$

- Wick's theorem at the level of observables.
- The coefficients $G(x-y)^j$ carry all singularities; the observables on the right have regular limits as $y \to x$.

Locality and factorisation

- O^1, O^2 with germs supported in $U^1_{1/t}$, $U^2_{1/t}$, $\operatorname{dist}(U^1, U^2) > 0$: the covariation $\langle DO^1_t, \dot{Q}_t\, DO^2_t \rangle \mathrm{d}t$ sees $\dot{Q}_t$ only between the supports.
- Finite range decomposition: it vanishes for large t, so the product exists and is asymptotically the pointwise product. Heat profile: exponentially small.
- Observables on disjoint regions multiply freely; products at the same point need renormalisation.

Factorisation algebra

This is the structure of a factorisation algebra of observables (precise axioms not established here).

Bauerschmidt–Brydges–Slade, LNM 2242 (finite range decompositions)

A semigroup of renormalisation maps

$\mathcal{R}_S(V) = -\log(\mu^S * \mathrm{e}^{-V})$. Since $\mu^S * \mu^{S'} = \mu^{S+S'}$,

$$\mathcal{R}_S \circ \mathcal{R}_{S'} = \mathcal{R}_{S+S'} = \mathcal{R}_{S'} \circ \mathcal{R}_S, \qquad V_t = \mathcal{R}_{Q_{T,t}}(V_T) :$$

an orbit of an abelian semigroup indexed by the cone of positive covariances, nonlinear on V, linear on e^{-V}. Generators:

$$\mathcal{P}_B(V) = \tfrac{1}{2}\,\mathrm{Tr}(B D^2 V) - \tfrac{1}{2}\langle DV, BDV\rangle, \qquad \partial_t V_t = -\mathcal{P}_{\dot{Q}_t}(V_t).$$

Proposition (Flatness)

(formal) $[\mathcal{P}_B, \mathcal{P}_{B'}] = 0$. *$V_t$ depends on the profile only through $Q_{T,t}$; $V_0(0)$ and ν_T only through Q_T.*

Different profiles with the same endpoints are different coordinate systems on the same RG trajectory: they change Φ and intermediate observables, not the terminal theory. No holonomy.

Perturbation theory and counterterms

Gaussian convolution lowers the Wick ordering: $V_t = \int [\lambda{:}\phi^4{:}_t + \gamma_T{:}\phi^2{:}_t] + O(\lambda^2)$, the dual of $\int H_4(X_t; c_t)$ being an observable. At second order, $S = Q_{T,t}$:

$$V_t = \int [\lambda{:}\phi^4{:}_t + \gamma_T{:}\phi^2{:}_t] - \frac{\lambda^2}{2} \sum_{k=1}^{4} \binom{4}{k}^2 k! \iint S(x,y)^k \, {:}\phi^{4-k}{:}_t(x) \, {:}\phi^{4-k}{:}_t(y) + O(\lambda^3).$$

$T \to \infty$ at fixed t, $d = 3$, $G(z) \sim (4\pi|z|)^{-1}$:

- $k = 1, 2$: finite, nonlocal.
- $k = 3$ (sunset): $S^3 \sim |z|^{-3}$; local divergence $-48\lambda^2 (\int S^3) \int {:}\phi^2{:}_t$, cancelled by $\gamma_T = 48\lambda^2 \int Q_T^3 = \frac{3\lambda^2}{\pi^2} \log T + O(1)$.
- $k = 4$: vacuum energy $\sim T$; cancels in ν_T.

Renormalisation = choosing V_T so that V_t at fixed resolution converges. $\nu_t \ll \mu^{Q_t}$ expected for $t < \infty$; the terminal Φ^4_3 is singular w.r.t. the GFF.

Dyson–Schwinger equations at every scale

Operators functions of A; $K = Q_T^{-1}$, $J_t = KQ_t$ (heat profile: a heat semigroup, $\to \mathrm{e}^{-A/t^2}$).

Theorem

(formal) The noise observable $\eta_t = K\Phi_t - J_t F_t$ *is a centred martingale,* $\mathrm{d}\eta_t = K\dot{Q}_t^{1/2}\mathrm{d}W_t^{\mathbb{P}} - J_t\,\mathrm{d}F_t$, *and*

$$\mathbb{E}^{\mathbb{P}}\left[D_{J_t h}G(\Phi_t)\right] = \mathbb{E}^{\mathbb{P}}\left[G(\Phi_t)\,\langle h, K\Phi_t - J_t F_t\rangle\right].$$

At $t = T$*: the DS equation* $\mathbb{E}^{\nu_T}[D_h G] = \mathbb{E}^{\nu_T}[G\langle h, K\varphi + DV_T\rangle]$.

Proof. $\dot{J}_t = K\dot{Q}_t$: drifts cancel. IBP for μ^{Q_t} in direction $J_t h$, using $\Phi_t \sim \nu_t$. □

The field equation at resolution t

As $T \to \infty$, $K \to A$, and $\eta_t = K\Phi_t - J_t F_t$ becomes

$$A\Phi_t = \mathrm{e}^{-A/t^2} F_t + \eta_t :$$

the Euclidean field equation $A\varphi = -V'(\varphi) + \text{noise}$, seen at resolution t. No products of distributions: the nonlinearity enters only through the force observable, smeared at scale $1/t$. For Φ^4_3, $t \to \infty$ gives the renormalised DS equation (Gubinelli–Hofmanová 2021).

Compare: stochastic quantisation (Lecture 2)

There the field equation is imposed in an auxiliary time; here it holds at each resolution, and the auxiliary parameter is the resolution itself.

Conditioning an SPDE solution on a scale filtration gives identities of the same shape but with a different noise: the martingale $J_t \, \mathrm{d}F_t$ in $\mathrm{d}\eta_t$ is essential.

Summary

- The GFF is the terminal value of a martingale X_t of smooth fields; observables are martingales, possibly without terminal value.
- Interactions: effective potentials V_t solve Polchinski; the interacting field solves an FBSDE whose force is an observable.
- In the forecast coordinate $\psi_t = \mathbb{E}_t \Phi_T$ the FBSDE has no drift, trees are integrated exactly, and the force flow is pure diffusion; in the gradient case it is Wetterich's equation.
- PDE (Polchinski) $\leftrightarrow$ dynamics (FBSDE) $\leftrightarrow$ variational principle (Boué–Dupuis).
- Products, OPE, counterterms and DS equations all live at finite resolution.

Next

Later lectures (not yet settled): probably a construction of Φ^4_3 on the torus via the FBSDE.

Exercises and reading

Exercises (numbering of the notes)

- 4.1 The field at resolution t; 4.2 divergence of c_t.
- 5.1 Quadratic interactions; 5.2 zero dimensions.
- 6.1 Stochastic maximum principle; 6.2 characteristics.
- 6.5 The constant; 6.7 1PI effective action.
- 6.8 Trees versus loops; 6.9 regulators.
- 7.2 Cost of the leading drift.
- 8.2 Products of Wick observables; 8.3 exponentials.
- 9.2 The sunset; 9.3 power counting.
- 10.1 DS from the dynamics.

Reading

- Bauerschmidt–Bodineau–Dagallier, *Stochastic dynamics and the Polchinski equation*, Probab. Surveys 2024.
- Barashkov–Gubinelli, *A variational method for* Φ^4_3, Duke 2020.
- Berges–Tetradis–Wetterich, Phys. Rep. 2002.
- Ma–Yong, *FBSDEs and their applications*, LNM 1702.
- Bailleul–Chevyrev–Gubinelli, *Stochastic observables and Wilson–Itô diffusions*, in prep.